

From NVDS and BiDAStabilizer to Safe Causal Temporal Stereo Refinement

Literature Review, Technical Analysis, and ARGOS Research Positioning

Leonardo Pampaloni
ARGOS Project
Aalborg University / Technical University of Denmark

July 2026 — Living Technical Document

Abstract

This document formalises the methodological positioning of the ARGOS temporal stereo refinement project.

The closest conceptual ancestor is the Neural Video Depth Stabilizer (NVDS), which introduced plug-and-play RGB-depth stabilisation for monocular video depth. The closest direct methodological baseline is BiDAStabilizer, a lightweight post-hoc stereo disparity refiner introduced in the TPAMI 2026 extension of BiDAStereo.

BiDAStabilizer demonstrates that explicit optical-flow alignment, local disparity feature extraction, and recurrent feature propagation can strongly improve temporal consistency. However, it is bidirectional, requires future frames, operates offline, is trained on generic imagery, and produces unbounded residual corrections.

ARGOS targets the remaining gap: a causal, online, plug-and-play, metric, safety-aware temporal stereo refiner for deformable surgical scenes.

The proposed direction is a forward-only causal adaptation of the BiDAStabilizer principle, augmented with target-to-source flow alignment, reliability masking, persistent aligned state, bounded gated disparity corrections, raw-good preservation, and strict out-of-distribution evaluation.

Contents

1	Executive Summary	4
2	Problem Definition	4
2.1	Input and Frozen Stereo Predictor	4
2.2	Desired Refiner	5
2.3	Metric Relationship	5
3	Taxonomy of Temporal Depth and Stereo Methods	5
3.1	Taxonomy Axes	5
4	Optical-Flow and Warping Conventions	7
4.1	Target-to-Source Flow	7
4.2	Warp-Support Mask	7
5	NVDS and the Plug-and-Play Stabilisation Paradigm	7
6	Temporal Stereo	8
6.1	Why Framewise Stereo Flickers	8
6.2	End-to-End Temporal Stereo Models	8

7	BiDAStabilizer: Closest Direct Baseline	9
7.1	BiDAStereo vs. BiDAStabilizer	9
7.2	Frozen Per-Frame Stereo Backbone	9
7.3	Local Disparity Alignment	9
7.4	Forward Propagation	10
7.5	Backward Propagation	10
7.6	Fusion and Residual Output	10
7.7	Why It Is Not Causal	10
8	BiDAStabilizer Training	10
8.1	Spatial Loss	10
8.2	Temporal Consistency Loss	11
8.3	Reported Configuration	11
9	BiDAStabilizer Ablation Evidence	11
9.1	Component Ablation	11
9.2	Propagation vs. Generic Temporal Aggregation	12
9.3	Local Before Global	12
9.4	Accuracy–Consistency Trade-Off	12
10	Surgical Depth and Stereo Context	13
10.1	Monocular Surgical Depth	13
10.2	Surgical Stereo	13
11	Gap Analysis	13
11.1	No Published Causal Surgical Stereo Refiner	13
11.2	No Explicit Metric-Safety Mechanism	14
11.3	No Standard Surgical Temporal Benchmark	14
12	ARGOS Research Hypothesis	14
13	Proposed Safe Causal BiDAStabilizer	14
13.1	High-Level Design	14
13.2	Alignment of Previous Evidence	15
13.3	Reliability Mask	15
13.4	Local Feature Encoder	15
13.5	Forward State Propagation	16
13.6	Safe Bounded Output	16
13.7	Target Model Size	16
14	Training Objective	16
14.1	Spatial Loss	16
14.2	Temporal Gradient Matching	16
14.3	Warp Consistency	17
14.4	Raw-Good Preservation	17
14.5	Correction Sparsity	17
15	Current ARGOS Experimental Context	17
15.1	Completed Experiments	17
15.2	NVDS-Lite Pilot	18
16	Baseline Ladder	18
17	Required Ablations	19

18 Correctness Tests	19
18.1 Flow-Direction Test	19
18.2 Mask Test	19
18.3 Functional Causal Leakage Test	20
18.4 Gradient and Graph Audit	20
18.5 State-Propagation Test	20
19 Evaluation Protocol	21
19.1 Dataset Roles	21
19.2 Geometric Metrics	21
19.3 Temporal Metrics with Processed Ground Truth	21
19.4 Prediction-Space Temporal Metrics	22
19.5 Safety Metrics	22
20 Decision Gates	22
21 Recommended Experimental Sequence	23
21.1 One-Seed Architectural Ladder	23
21.2 Promotion to Three Seeds	23
21.3 OOD Evaluation	24
22 Risks and Alternative Outcomes	24
22.1 Causal–Bidirectional Gap	24
22.2 Identity Collapse	24
22.3 Temporal Over-Smoothing	24
22.4 Domain Overfitting	25
22.5 Insufficient Data	25
23 Two-Month Research Roadmap	25
23.1 Weeks 1–2	25
23.2 Weeks 3–4	25
23.3 Weeks 5–6	25
23.4 Weeks 7–8	26
24 Final Positioning	26
25 Verification Notes	26

1 Executive Summary

The ARGOS objective is to stabilise the temporally inconsistent disparity produced by a frozen image-based stereo predictor, while preserving metric accuracy and avoiding unsafe corrections.

The main methodological hierarchy is:

1. **NVDS / NVDS+** established the plug-and-play stabilisation paradigm for monocular video depth using RGB-depth features and local cross-frame reasoning [1, 2].
2. **DynamicStereo, TemporalStereo, TC-Stereo, BiDAStereo, Stereo Any Video, and PPMStereo** developed increasingly sophisticated temporal stereo architectures [5–8, 10?].
3. **BiDAStabilizer** is the closest direct baseline for ARGOS: it is a lightweight, plug-and-play stereo disparity refiner operating on top of a frozen image-based stereo model [9].
4. **Surgical temporal depth methods**, including Surgical-DINO, EndoDAV, EndoStreamDepth, and temporally supervised surgical monocular depth, establish the domain context but do not solve the same causal, plug-and-play, metric stereo problem [13? –15].

The core literature-derived conclusion is:

Main Architectural Conclusion

The most promising ARGOS architecture is not a generic temporal attention module. The strongest direct evidence supports the following sequence:

aligned local evidence $\longrightarrow$ forward propagated state $\longrightarrow$ safety-bounded metric residual.

This corresponds to a causal, surgical, safety-aware adaptation of BiDAStabilizer rather than a larger NVDS-style attention model.

A qualified novelty statement is therefore:

To the best of our knowledge, the reviewed literature contains no published causal, plug-and-play temporal disparity refiner designed and comprehensively evaluated for deformable surgical stereo video with explicit metric-safety constraints.

This claim is literature-based and should be revised if new evidence emerges.

2 Problem Definition

2.1 Input and Frozen Stereo Predictor

Consider a calibrated stereo endoscope producing a sequence

$$\left\{ \left(I_L^t, I_R^t \right) \right\}_{t=1}^T.$$

A frozen image-based stereo predictor $\mathcal{S}$ produces a raw disparity map independently for each frame:

$$D_{\text{raw}}^t = \mathcal{S} \left(I_L^t, I_R^t \right).$$

Because each frame is processed independently, small changes in appearance, specularity, occlusion, deformation, and texture may cause strong frame-to-frame disparity fluctuations.

2.2 Desired Refiner

A temporal refiner $\mathcal{R}$ should produce

$$D_{\text{ref}}^t = \mathcal{R} \left(I_L^{\leq t}, D_{\text{raw}}^{\leq t} \right),$$

subject to the following constraints:

1. **Causality**: only current and past frames may be used.
2. **Metric preservation**: output remains calibrated disparity.
3. **Temporal improvement**: flickering and isolated corrections are reduced.
4. **Spatial safety**: already-correct raw predictions are protected.
5. **Bounded output**: corrections cannot grow without limit.
6. **OOD robustness**: improvements should persist outside the SCARED development domain.
7. **Online deployment**: runtime should be compatible with surgical robotic use.

2.3 Metric Relationship

For calibrated stereo,

$$Z = \frac{fb}{d},$$

where Z is depth, f is focal length, b is stereo baseline, and d is disparity.

A small disparity error can induce a large depth error at low disparity. Therefore, temporal smoothing cannot be judged independently from metric geometry.

3 Taxonomy of Temporal Depth and Stereo Methods

3.1 Taxonomy Axes

Methods are classified according to:

1. monocular or stereo;
2. end-to-end estimator or post-hoc refiner;
3. causal or bidirectional;
4. local temporal window or persistent/global memory;
5. unaligned fusion, learned correspondence, or explicit geometric alignment;
6. relative or metric output;
7. generic or surgical imagery;
8. unconstrained or bounded correction.

Table 1: Taxonomy of relevant temporal depth and disparity methods.

Method	M/S	E/R	C/B	L/G	Temporal mecha- nism	Rel./Mon.	Gen./Surg.	Free/Bnd.	Venue
NVDS [1]	M	R	B	L	learned corre- spon- dence	Rel.	Gen.	Free	ICCV 2023
NVDS+ [2]	M	R	B	L	learned corre- spon- dence	Rel.	Gen.	Free	TPAMI 2024
Video Depth Anything [3]	M	E	C	G	temporal atten- tion	Rel.	Gen.	Free	CVPR 2025
FlashDepth [4]	M	E	C	G	state- space memory	Rel.	Gen.	Free	ICCV 2025
DynamicStereo [5]	S	E	B	L	stereo- temporal atten- tion	Met.	Gen.	Free	CVPR 2023
TemporalStereo [6]	S	E	C	L	aligned geome- try reuse	Met.	Gen.	Free	IROS 2023
TC-Stereo [7]	S	E	C	G	pose align- ment + state	Met.	Gen.	Free	ECCV 2024
BiDAStereo [8]	S	E	B	G	flow align- ment + propa- gation	Met.	Gen.	Free	ECCV 2024
BiDAStabilizer [9]	S	R	B	G	flow align- ment + propa- gation	Met.	Gen.	Free	TPAMI 2026
Stereo Any Video [10]	S	E	B	G	monocular tempo- ral priors	Met.	Gen.	Free	ICCV 2025
PPMStereo [11]	S	E	B	G	selective tempo- ral memory	Met.	Gen.	Free	NeurIPS 2025
StereoDiffusion [12]	S	E	?	L	flow- aligned diffu- sion prior	Met.	Surg.	Free	MICCAI 2024
EndoDAV [14]	M	E	C	G	adapted video- depth prior	Rel.	Surg.	Free	MICCAI 2025
EndoStreamDepth [15]	M	E	C	G	hierarchical Mamba	Rel.	Surg.	Free	MIDL 2026
ARGOS NVDS-lite pilot	S	R	C	L	local correla- tion	Met. 6	Surg.	Bnd.	ongoing
ARGOS causal	S	R	C	G	aligned	Met.	Surg.	Bnd.	proposed

4 Optical-Flow and Warping Conventions

4.1 Target-to-Source Flow

To eliminate ambiguity, this document avoids the labels “forward flow” and “backward flow” when defining the warping operation.

Let

$$\mathbf{f}_{t \rightarrow t-1} = \mathbf{f}_{t \rightarrow t-1}$$

denote the optical flow evaluated in the target frame t that points toward the corresponding source coordinates in frame $t - 1$.

A grid-sampling warp is defined as

$$\mathcal{W}\left(X^{t-1}, \mathbf{f}_{t \rightarrow t-1}\right)(\mathbf{p}) = X^{t-1}(\mathbf{p} + \mathbf{f}_{t \rightarrow t-1}(\mathbf{p})).$$

Therefore, to warp information from frame $t - 1$ into the coordinate system of frame t , the required sampling field is the target-to-source flow $\mathbf{f}_{t \rightarrow t-1}$.

Implementation Rule

If a warping function samples a source tensor at $\mathbf{p} + \mathbf{f}(\mathbf{p})$, the input flow must map target coordinates to source coordinates.

Using a source-to-target flow directly in such a function produces the wrong warp direction.

4.2 Warp-Support Mask

The valid support for a warped previous-frame quantity is

$$M^t = M_{\text{valid}}^t \cdot \mathcal{W}\left(M_{\text{valid}}^{t-1}, \mathbf{f}_{t \rightarrow t-1}\right) \cdot M_{\text{inb}}^t \cdot M_{\text{occ}}^t \cdot M_{\text{finite}}^t.$$

Importantly, the current validity mask must never be intersected with the *unwarped* previous-frame validity mask, because the two masks are defined in different coordinate systems.

5 NVDS and the Plug-and-Play Stabilisation Paradigm

NVDS introduced a generic post-hoc stabilisation network for monocular video depth [1].

The model receives:

- RGB frames;
- initial flickering disparity or depth predictions;
- a target frame;
- three local reference frames.

RGB and depth are encoded into depth-aware features, and local cross-attention is used to relate the target frame to references within a sliding temporal window.

NVDS is trained using a combination of:

- spatial affinity-invariant losses;
- gradient matching;
- optical-flow-based temporal consistency.

The main lessons inherited by ARGOS are:

1. freeze the upstream estimator;
2. refine rather than re-estimate from scratch;
3. combine appearance and geometry;
4. use explicit temporal supervision;
5. evaluate temporal and spatial quality jointly.

However, NVDS is not a direct solution for ARGOS because it is:

- monocular;
- primarily relative-depth based;
- bidirectional at inference;
- trained on natural scenes;
- based on a short local window;
- unconstrained in its output correction.

6 Temporal Stereo

6.1 Why Framewise Stereo Flickers

Even metric stereo models can flicker temporally because:

1. cost volumes are recomputed independently for every frame;
2. small changes in specularity modify the matching landscape;
3. textureless and occluded regions remain underconstrained;
4. recurrent stereo models often reinitialise their hidden state;
5. non-rigid tissue violates simple rigid temporal assumptions;
6. surgical tools create rapidly appearing and disappearing occlusions.

6.2 End-to-End Temporal Stereo Models

DynamicStereo uses divided temporal and stereo attention over video windows [5].

TemporalStereo reuses past geometric and contextual evidence and supports single-frame fall-back [6].

TC-Stereo uses warped previous disparity and recurrent temporal state, but depends on camera poses for temporal projection [7].

BiDAStereo combines:

- bidirectional optical-flow alignment;
- triple-frame correlation;
- motion-propagation recurrent units;
- global temporal consistency.

PPMStereo introduces a selective pick-and-play memory mechanism for longer temporal horizons [11].

Stereo Any Video injects frozen temporally stable monocular priors into a stereo framework [10].

These methods are important upper bounds and architectural references, but most are complete end-to-end stereo systems rather than plug-and-play refiners.

7 BiDAStabilizer: Closest Direct Baseline

7.1 BiDAStereo vs. BiDAStabilizer

The TPAMI extension of “Match Stereo Videos via Bidirectional Alignment” contains two distinct systems [9]:

Table 2: BiDAStereo and BiDAStabilizer are separate systems.

Property	BiDAStereo	BiDAStabilizer
System type	Complete temporal stereo model	Post-hoc disparity stabiliser
Input	Stereo video	Disparities from a frozen stereo backbone
Stereo backbone	Internal	External and frozen
Trainable parameters	Approximately 12.2 M	Approximately 0.7 M
Training iterations	Approximately 100 K	Approximately 8 K
Training time	Approximately 5 days	Approximately 0.5 day
Output	Full disparity estimate	Residual correction added to input disparity
Temporal direction	Bidirectional	Bidirectional

BiDAStabilizer is therefore the closest published stereo analogue of NVDS.

7.2 Frozen Per-Frame Stereo Backbone

For every frame, a frozen image-based stereo model produces

$$d^t = \mathcal{S}(I_L^t, I_R^t).$$

The paper demonstrates the stabiliser using:

- RAFT-Stereo;
- IGEV-Stereo.

The stereo backbone remains frozen during stabiliser training.

7.3 Local Disparity Alignment

The previous and next disparities are aligned to the current frame:

$$\hat{d}^{t-1} = \mathcal{W}(d^{t-1}, \mathbf{f}_{t \rightarrow t-1}),$$

$$\hat{d}^{t+1} = \mathcal{W}(d^{t+1}, \mathbf{f}_{t \rightarrow t+1}).$$

A local convolutional feature extractor produces

$$F_d^t = \text{FE}(\hat{d}^{t-1}, d^t, \hat{d}^{t+1}).$$

This stage creates local temporally aligned disparity evidence.

7.4 Forward Propagation

The forward hidden state is initialised as

$$F_f^0 = \mathbf{0}.$$

At each frame,

$$\hat{F}_f^{t-1} = \mathcal{W}\left(F_f^{t-1}, \mathbf{f}_{t \rightarrow t-1}\right),$$

$$F_f^t = \text{PropEnc}\left(\hat{F}_f^{t-1}, F_d^t\right).$$

The Propagation Encoder is a shared convolutional encoder. It is not described as a GRU in the BiDAStabilizer formulation.

7.5 Backward Propagation

Similarly,

$$F_b^T = \mathbf{0},$$

$$\hat{F}_b^{t+1} = \mathcal{W}\left(F_b^{t+1}, \mathbf{f}_{t \rightarrow t+1}\right),$$

$$F_b^t = \text{PropEnc}\left(\hat{F}_b^{t+1}, F_d^t\right).$$

7.6 Fusion and Residual Output

Forward and backward hidden states are fused:

$$\Delta d^t = \text{Fusion}\left(F_f^t, F_b^t\right),$$

and the final disparity is

$$d_{\text{out}}^t = d^t + \Delta d^t.$$

BiDAStabilizer predicts one residual per frame and does not perform iterative disparity refinement.

7.7 Why It Is Not Causal

BiDAStabilizer is bidirectional in two separate ways:

1. local features use both d^{t-1} and d^{t+1} ;
2. the model performs both forward and backward hidden-state propagation.

Consequently, it requires future frames and effectively operates as an offline or batch model. This prevents direct deployment in an online surgical robotic control loop.

8 BiDAStabilizer Training

8.1 Spatial Loss

For one output disparity per frame, the spatial loss reduces to

$$\mathcal{L}_s = \sum_t \left\| d_{\text{gt}}^t - d_{\text{out}}^t \right\|_1.$$

8.2 Temporal Consistency Loss

The aligned temporal consistency loss compares the current output to aligned neighbouring outputs:

$$\mathcal{L}_c = \sum_t \left[O_{t-1} \left\| \mathcal{W} \left(d_{\text{out}}^{t-1}, \mathbf{f}_{t \rightarrow t-1} \right) - d_{\text{out}}^t \right\|_1 \right. \\ \left. + O_{t+1} \left\| \mathcal{W} \left(d_{\text{out}}^{t+1}, \mathbf{f}_{t \rightarrow t+1} \right) - d_{\text{out}}^t \right\|_1 \right],$$

where O denotes flow-derived occlusion support.

The total objective is

$$\mathcal{L} = \mathcal{L}_s + \lambda \mathcal{L}_c,$$

with

$$\lambda = 0.2$$

for BiDAStabilizer.

8.3 Reported Configuration

The reported training setup includes:

Table 3: Reported BiDAStabilizer training configuration.

Parameter	Value
Sequence length	$T = 8$
Training iterations	8 000
Resolution	256×512
Optimiser	AdamW
Initial learning rate	4×10^{-4}
Schedule	One-cycle
Temporal loss weight	$\lambda = 0.2$
Trainable parameters	approximately 0.7 M
Training time	approximately 0.5 day on A100
Batch size	not explicitly stated for BiDAStabilizer

Primary training data include SceneFlow and Dynamic Replica, with robustness variants incorporating KITTI Depth and Infinigen SV.

9 BiDAStabilizer Ablation Evidence

9.1 Component Ablation

Table 4: BiDAStabilizer component ablation. Lower is better.

Setup	Sintel δ_{1px}^t	Sintel TEPE	DR δ_{1px}^t	DR TEPE
Full model	12.31	1.73	0.60	0.046
Without alignment	13.43	1.88	0.81	0.070
Without propagation	13.18	1.85	0.79	0.072
Without feature-based fusion	13.10	1.79	0.67	0.055

Removing alignment increases Dynamic Replica TEPE by approximately 52%.

Removing propagation increases Dynamic Replica TEPE by approximately 57%.

Therefore, both components are central rather than auxiliary.

9.2 Propagation vs. Generic Temporal Aggregation

Table 5: Propagation compared with generic temporal aggregation alternatives.

Temporal mechanism	Sintel $\delta_{1\text{px}}^t$	Sintel TEPE	DR $\delta_{1\text{px}}^t$	DR TEPE
3D UNet	21.21	2.23	7.99	0.358
3D convolution	14.69	1.96	1.26	0.117
1D attention	14.83	2.03	1.25	0.116
1D + 2D attention	14.51	1.97	0.94	0.088
3D attention		Out of memory		
BiDA propagation	12.31	1.73	0.60	0.046

The propagation architecture outperforms all tested attention and spatiotemporal convolution alternatives.

This is a central design signal for ARGOS.

9.3 Local Before Global

A related BiDAStereo ablation shows:

Table 6: Local matching must precede global propagation.

Local	Global	Sintel $\delta_{1\text{px}}^t$	Sintel TEPE	DR $\delta_{1\text{px}}^t$	DR TEPE
No	No	11.19	1.33	0.69	0.079
Yes	No	11.10	1.31	0.64	0.067
No	Yes	13.97	1.78	0.82	0.076
Yes	Yes	11.09	1.24	0.61	0.062

Global propagation without valid local evidence can degrade performance.
This suggests that the order

local alignment $\rightarrow$ local feature extraction $\rightarrow$ state propagation
is architecturally meaningful.

9.4 Accuracy–Consistency Trade-Off

BiDAStereo does not improve every spatial metric.

Reported examples include:

- small increases in Bad-3 or related spatial thresholds;
- occasional marginal degradation in OPW on KITTI;
- substantial spatial degradation under some mixed-domain training configurations.

Therefore, temporal consistency cannot be used as the only success criterion.

For surgical use, a temporal method must also report:

- new-bad pixels;
- harmful corrections;
- raw-good preservation;
- correction magnitude;
- OOD geometry;
- seed stability.

10 Surgical Depth and Stereo Context

10.1 Monocular Surgical Depth

Relevant surgical monocular approaches include:

- appearance-flow self-supervision for endoscopic imagery;
- Surgical-DINO foundation-model adaptation;
- EndoDAC;
- temporal fine-tuning of Depth Anything;
- EndoDAV;
- EndoStreamDepth.

These methods demonstrate the importance of surgical-domain adaptation and temporal modelling, but they do not provide a plug-and-play metric stereo stabiliser.

10.2 Surgical Stereo

StereoDiffusion uses optical-flow-warped previous disparity as a temporal prior for diffusion-based surgical stereo [12].

StereoMamba addresses robust intraoperative stereo matching but remains primarily a per-frame model [17].

The surgical setting remains difficult because of:

1. deformable tissue;
2. specular highlights;
3. non-Lambertian illumination;
4. blood, smoke, and fog;
5. tool occlusions;
6. low texture;
7. rapid endoscope motion;
8. sparse or processed ground truth.

11 Gap Analysis

The reviewed literature supports the following qualified gaps.

11.1 No Published Causal Surgical Stereo Refiner

A published causal, plug-and-play disparity refiner specifically designed for deformable surgical stereo video was not identified in the reviewed body of work.

NVDS is monocular.

BiDASStabilizer is stereo and plug-and-play but bidirectional and generic.

TC-Stereo is causal but is an end-to-end estimator and relies on camera pose.

Surgical methods are generally monocular or full estimators rather than post-hoc stereo refiners.

11.2 No Explicit Metric-Safety Mechanism

The reviewed temporal stereo models generally produce unconstrained disparity updates.

In a surgical setting, changing an already-correct metric prediction can be more dangerous than abstaining.

This motivates:

- bounded residuals;
- confidence gates;
- raw-good preservation losses;
- false-activation metrics;
- catastrophic-correction analysis.

11.3 No Standard Surgical Temporal Benchmark

SCARED provides official structured-light data at keyframes.

Any internally rectified, propagated, filtered, or densified temporal target must be described as *internal processed supervision*, not as native official dense temporal ground truth.

D4D provides full surgical clips and sparse Zivid anchors and is therefore suited to prediction-space temporal evaluation plus independent sparse geometry.

SERV-CT provides static CT-derived disparity and should be used only for static OOD geometry and safety.

12 ARGOS Research Hypothesis

The central hypothesis is:

ARGOS Hypothesis

A forward-only, plug-and-play temporal stereo refiner can improve the temporal consistency of a frozen surgical stereo predictor without materially degrading metric geometry, provided that past evidence and persistent state are explicitly aligned into the current frame and the output correction is bounded and confidence-gated.

This remains a hypothesis until validated on:

- SCARED;
- D4D;
- SERV-CT.

13 Proposed Safe Causal BiDAStabilizer

13.1 High-Level Design

The proposed model is a forward-only causal adaptation of BiDAStabilizer.

At time t , the model receives:

- current left RGB I_L^t ;
- current raw disparity D_{raw}^t ;
- previous left RGB I_L^{t-1} ;

- previous raw disparity D_{raw}^{t-1} ;
- previous hidden state H^{t-1} ;
- target-to-source flow $\mathbf{f}_{t \rightarrow t-1}$;
- current and previous validity masks;
- occlusion and flow-consistency masks.

No frame $t + 1$ is available.

13.2 Alignment of Previous Evidence

Previous disparity is aligned as

$$\tilde{D}_{\text{raw}}^{t-1} = \mathcal{W} \left(D_{\text{raw}}^{t-1}, \mathbf{f}_{t \rightarrow t-1} \right).$$

Previous RGB is aligned as

$$\tilde{I}^{t-1} = \mathcal{W} \left(I_L^{t-1}, \mathbf{f}_{t \rightarrow t-1} \right).$$

Previous hidden state is aligned as

$$\tilde{H}^{t-1} = \mathcal{W} \left(H^{t-1}, \mathbf{f}_{t \rightarrow t-1} \right).$$

Previous validity is also warped:

$$\tilde{M}_{\text{valid}}^{t-1} = \mathcal{W} \left(M_{\text{valid}}^{t-1}, \mathbf{f}_{t \rightarrow t-1} \right).$$

13.3 Reliability Mask

The alignment support is

$$M^t = M_{\text{valid}}^t \cdot \tilde{M}_{\text{valid}}^{t-1} \cdot M_{\text{inb}}^t \cdot M_{\text{-occ}}^t \cdot M_{\text{finite}}^t.$$

Optional future extensions may include:

- specularity confidence;
- stereo matching confidence;
- tool segmentation;
- flow uncertainty;
- photometric residual confidence.

13.4 Local Feature Encoder

A compact convolutional local encoder receives a feature stack such as

$$X_{\text{local}}^t = \left[D_{\text{raw}}^t, \tilde{D}_{\text{raw}}^{t-1}, D_{\text{raw}}^t - \tilde{D}_{\text{raw}}^{t-1}, \left| D_{\text{raw}}^t - \tilde{D}_{\text{raw}}^{t-1} \right|, I_L^t, \tilde{I}^{t-1}, M^t, \nabla D_{\text{raw}}^t \right].$$

It produces

$$F_{\text{local}}^t = \text{LocalEnc} \left(X_{\text{local}}^t \right).$$

The main candidate should use a lightweight convolutional encoder rather than a large local correlation volume.

The existing NVDS-lite local-correlation model remains a baseline.

13.5 Forward State Propagation

The persistent state update is

$$H^t = \text{PropEnc} \left(\tilde{H}^{t-1}, F_{\text{local}}^t, M^t \right).$$

The initial state is

$$H^0 = \mathbf{0}.$$

No backward temporal pass is performed.

No state may cross sequence boundaries.

A state-reset option must be exposed for ablation.

13.6 Safe Bounded Output

A fusion head predicts:

$$g^t \in [0, 1]$$

and an unconstrained residual logit δ^t .

The final disparity is

$$D_{\text{ref}}^t = D_{\text{raw}}^t + g^t r_{\text{max}} \tanh \left(\delta^t \right).$$

The applied correction is therefore bounded:

$$\left| D_{\text{ref}}^t - D_{\text{raw}}^t \right| \leq r_{\text{max}}.$$

The gate enables selective abstention:

$$g^t \approx 0 \implies D_{\text{ref}}^t \approx D_{\text{raw}}^t.$$

13.7 Target Model Size

The desired trainable parameter range is

$$0.5 \text{ M} \leq N_{\text{params}} \leq 2 \text{ M}.$$

The frozen stereo and optical-flow models are excluded from the trainable parameter count.

14 Training Objective

The proposed loss is

$$\mathcal{L} = \mathcal{L}_{\text{spatial}} + \lambda_{\text{TGM}} \mathcal{L}_{\text{TGM}} + \lambda_{\text{warp}} \mathcal{L}_{\text{warp}} + \lambda_{\text{safe}} \mathcal{L}_{\text{raw-good}} + \lambda_{\text{sparse}} \mathcal{L}_{\text{corr}}.$$

14.1 Spatial Loss

$$\mathcal{L}_{\text{spatial}} = \frac{\sum_{t, \mathbf{p}} M_{\text{valid}}^t(\mathbf{p}) \left| D_{\text{ref}}^t(\mathbf{p}) - D_{\text{gt}}^t(\mathbf{p}) \right|}{\sum_{t, \mathbf{p}} M_{\text{valid}}^t(\mathbf{p})}.$$

14.2 Temporal Gradient Matching

$$\mathcal{L}_{\text{TGM}} = \frac{1}{T-1} \sum_{t=2}^T \left| \left(D_{\text{ref}}^t - D_{\text{ref}}^{t-1} \right) - \left(D_{\text{gt}}^t - D_{\text{gt}}^{t-1} \right) \right|.$$

Internally propagated SCARED targets must be described as processed internal supervision rather than official dense temporal ground truth.

14.3 Warp Consistency

The previous refined disparity is warped into the current frame:

$$\tilde{D}_{\text{ref}}^{t-1} = \mathcal{W} \left(D_{\text{ref}}^{t-1}, \mathbf{f}_{t \rightarrow t-1} \right).$$

Then

$$\mathcal{L}_{\text{warp}} = \frac{\sum_{t, \mathbf{p}} M^t(\mathbf{p}) \left| D_{\text{ref}}^t(\mathbf{p}) - \tilde{D}_{\text{ref}}^{t-1}(\mathbf{p}) \right|}{\sum_{t, \mathbf{p}} M^t(\mathbf{p})}.$$

A literature-aligned initial value is

$$\lambda_{\text{warp}} = 0.2.$$

This follows the BiDASStabilizer temporal consistency weight and should be used as a starting point rather than treated as an optimised value.

14.4 Raw-Good Preservation

Define

$$M_{\text{good}}^t = M_{\text{valid}}^t \cdot \mathbb{I} \left[\left| D_{\text{raw}}^t - D_{\text{gt}}^t \right| < \tau_{\text{good}} \right].$$

Then

$$\mathcal{L}_{\text{raw-good}} = \frac{\sum_{t, \mathbf{p}} M_{\text{good}}^t(\mathbf{p}) \left| D_{\text{ref}}^t(\mathbf{p}) - D_{\text{raw}}^t(\mathbf{p}) \right|}{\sum_{t, \mathbf{p}} M_{\text{good}}^t(\mathbf{p})}.$$

14.5 Correction Sparsity

$$\mathcal{L}_{\text{corr}} = \frac{1}{T H W} \sum_{t, \mathbf{p}} \left| D_{\text{ref}}^t(\mathbf{p}) - D_{\text{raw}}^t(\mathbf{p}) \right|.$$

The objective is selective correction, not uniform smoothing.

15 Current ARGOS Experimental Context

15.1 Completed Experiments

The following have already been completed.

Frozen S2M2 baseline. S2M2 is used as the frozen framewise stereo predictor.

Disparity-only causal ConvGRU / VDPP-style model. A lightweight disparity-only recurrent model was trained with spatial and TGM supervision.

The corrected conclusion is:

- TGM acts as a useful temporal regulariser;
- full causal history does not robustly beat shuffled history;
- the architecture does not demonstrate reliable temporal-history use;
- D4D transfer remains absent or marginal.

The result is classified as **marginal**, not confirmed.

15.2 NVDS-Lite Pilot

A corrected NVDS-lite pilot has included:

- RGB-disparity encoding;
- local target-to-history correlation;
- bounded gated residual;
- target-to-source optical-flow warping;
- warped previous validity;
- explicit in-bounds support;
- forward–backward flow consistency;
- future-frame perturbation tests;
- gradient and computation-graph audits.

The pilot remains useful as a local-correlation baseline.

However, the literature review changes its role:

NVDS-lite should no longer be treated as the primary final candidate. It should be retained as the local, non-propagating baseline in a broader aligned-local-to-propagation ladder.

16 Baseline Ladder

Table 7: Recommended ARGOS baseline ladder.

#	Configuration	Scientific purpose
0	Frozen raw S2M2	No temporal refinement
1	Current-frame-only bounded refiner	Spatial correction without history
2	Naive causal multi-frame concatenation	Unaligned temporal fusion
3	Historical disparity-only ConvGRU	Previous ARGOS recurrent baseline
4	NVDS-lite local correlation	Learned local correspondence without persistent state
5	Aligned-local-only convolutional refiner	Explicit flow alignment without memory
6	Causal BiDA-style propagation with unconstrained residual	Isolate value of forward propagated state
7	Causal BiDA-style propagation with bounded gate	Add metric-safety constraint
8	Full ARGOS model with all reliability masks	Complete safe causal candidate
9	Offline bidirectional BiDA-style upper bound	Estimate causal–bidirectional gap

The intended architectural sequence is

current only $\rightarrow$ unaligned temporal $\rightarrow$ aligned local $\rightarrow$ aligned local + persistent state $\rightarrow$ safe persistent state.

17 Required Ablations

Table 8: Required ablation experiments.

Ablation	Question
No alignment	Does explicit flow-based correspondence matter?
No persistent state	Does propagation add value beyond local evidence?
No RGB	Does appearance improve disparity-only reasoning?
No reliability mask	Are occlusion, in-bounds, and warped-valid masks necessary?
No safety bound	Does bounded correction prevent harmful updates?
No warp loss	Does explicit motion-compensated supervision help?
No TGM	Does temporal-gradient supervision add complementary value?
Shuffled history	Does correct temporal history matter?
Current-only	Does history help at all?
State reset	Does persistent long-range state matter?
One vs. three past frames	What temporal horizon is required?
SCARED-only vs. multi-dataset	Does training diversity improve transfer?

18 Correctness Tests

Before meaningful training, the following tests are mandatory.

18.1 Flow-Direction Test

On at least 20 consecutive SCARED frame pairs:

1. warp previous RGB with $\mathbf{f}_{t \rightarrow t-1}$;
2. test the opposite direction;
3. compare masked photometric error;
4. compare against no warp;
5. inspect contact sheets.

The chosen direction must agree with:

- the mathematical grid-sampling convention;
- the empirical photometric result.

18.2 Mask Test

The mask validation must report:

- warped previous validity;
- in-bounds support;
- forward-backward consistency;
- occluded fraction;
- out-of-bounds fraction;
- finite-value support;
- NaN and Inf counts.

18.3 Functional Causal Leakage Test

For any time t , perturb all frames after t :

$$I^\tau \leftarrow \tilde{I}^\tau, \quad D^\tau \leftarrow \tilde{D}^\tau, \quad \tau > t.$$

Then verify

$$\max_{\tau \leq t} \left| D_{\text{ref}_{\text{original}}}^\tau - D_{\text{ref}_{\text{perturbed}}}^\tau \right| < \epsilon.$$

Checking only history indices is insufficient; the output itself must remain numerically unchanged.

18.4 Gradient and Graph Audit

The audit must verify:

1. every trainable refiner parameter receives a finite gradient;
2. no RAFT parameter receives gradients;
3. no S2M2 parameter receives gradients;
4. all expected loss terms are finite;
5. expected losses are nonzero where valid support exists.

18.5 State-Propagation Test

The model must also verify:

- perturbing H^{t-1} changes H^t ;
- resetting H^{t-1} reproduces the state-reset baseline;
- state does not cross sequence boundaries;
- invalid support suppresses unreliable state injection.

19 Evaluation Protocol

19.1 Dataset Roles

Table 9: ARGOS dataset roles.

Dataset	Training use	Geometry supervision	Temporal signal	Role
SCARED [18]	Source-domain development	Official structured-light keyframes plus internal processed targets	Internal processed temporal supervision	Training and validation
D4D [19]	Never train or tune	Sparse Zivid anchors	Full surgical clips	Main OOD temporal and sparse geometry
SERV-CT [20]	Never train or tune	Static CT-derived disparity	None	Static OOD geometry and safety
Hamlyn	Optional diversity	Stereo-derived pseudo-ground-truth	Surgical videos	Optional pretraining
EndoSLAM [21]	Optional diversity	Scanner and trajectory information	Endoscopic videos	Optional pretraining
StereoMIS [22]	Qualitative only	No dense disparity ground truth	Stereo video	Prediction-space qualitative analysis

19.2 Geometric Metrics

Report:

- MAE or EPE;
- Bad-1, Bad-3, Bad-5;
- new-Bad-1 and new-Bad-3;
- harmful and beneficial correction rates;
- raw-good preservation;
- boundary error;
- modified-pixel ratio;
- correction magnitude percentiles.

19.3 Temporal Metrics with Processed Ground Truth

Report:

- TGM error;
- temporal end-point or stability error;
- temporal error jitter;
- high-frequency error energy;
- boundary temporal error.

These metrics must be labelled as based on internally processed temporal targets when official dense temporal ground truth is unavailable.

19.4 Prediction-Space Temporal Metrics

Report:

- motion-compensated disparity inconsistency;
- correction flicker;
- sign-flip rate;
- isolated activation;
- valid warp support;
- low-motion and high-motion strata;
- persistent-state benefit;
- state-reset degradation.

19.5 Safety Metrics

Report:

- false activation on raw-good pixels;
- catastrophic harmful corrections;
- correction magnitude distribution;
- correction sparsity;
- per-seed variance;
- OOD sign stability.

20 Decision Gates

The full model advances only if all relevant gates hold.

1. Aligned-local-only beats naive unaligned concatenation.
2. Forward propagation beats aligned-local-only.
3. Full causal model beats current-only.
4. Full causal model beats shuffled history.
5. Persistent state beats state reset.
6. Correction remains nonzero but bounded.
7. Raw-good preservation remains acceptable.
8. SCARED geometry does not materially degrade.
9. D4D temporal improvement has stable sign.
10. D4D sparse anchors remain geometrically safe.
11. SERV-CT shows no major false activation.

12. Inference latency is compatible with online surgical use.

A target runtime is

$$t_{\text{frame}} \leq 100 \text{ ms}$$

at the selected evaluation resolution.

Falsification Rule

If the model improves SCARED temporal metrics but degrades D4D or SERV-CT, the result must be classified as source-domain overfitting rather than success.

21 Recommended Experimental Sequence

21.1 One-Seed Architectural Ladder

Before a full multi-seed matrix, run one seed for:

1. raw S2M2;
2. current-only bounded refiner;
3. naive causal concatenation;
4. NVDS-lite local correlation;
5. aligned-local-only convolutional model;
6. causal BiDA propagation without safety gate;
7. full safe causal BiDA model;
8. shuffled-history full model;
9. state-reset full model.

This one-seed ladder tests the architectural sequence before expensive replication.

21.2 Promotion to Three Seeds

Only decisive configurations should be promoted to three seeds.

At minimum:

- aligned-local-only;
- full causal propagation;
- current-only;
- shuffled-history;
- state-reset;
- unaligned concatenation;
- raw baseline.

21.3 OOD Evaluation

After promotion:

- evaluate all promoted models on D4D;
- evaluate all promoted models on SERV-CT;
- never tune using D4D or SERV-CT;
- report paired bootstrap confidence intervals against raw S2M2.

22 Risks and Alternative Outcomes

22.1 Causal–Bidirectional Gap

BiDAStabilizer benefits from:

- future disparity;
- future-aligned local evidence;
- backward sequence propagation.

A causal version may underperform substantially.

If so, a valid contribution may be:

A quantified causal–bidirectional performance gap for temporal disparity stabilisation in deformable surgical stereo.

22.2 Identity Collapse

Strong safety and sparsity regularisation may force

$$D_{\text{ref}}^t \rightarrow D_{\text{raw}}^t.$$

This must be detected using:

- modified-pixel ratio;
- applied correction magnitude;
- gate distribution;
- seed stability.

A model that collapses to identity is safe but scientifically uninformative as a temporal refiner.

22.3 Temporal Over-Smoothing

A temporal model may reduce flicker by suppressing legitimate motion.

Mitigation requires:

- TGM or GT-relative temporal metrics;
- high-motion strata;
- motion-lag proxies;
- boundary temporal accuracy;
- geometry preservation.

22.4 Domain Overfitting

A model may learn processed SCARED target artefacts rather than transferable surgical correspondence.

This is why D4D and SERV-CT are hard decision gates rather than optional additional experiments.

22.5 Insufficient Data

If the correct architecture uses temporal evidence but transfers poorly, the next step should be:

- multi-dataset pretraining;
- Hamlyn or EndoSLAM diversity;
- surgery-adapted foundation features;
- confidence-aware adaptation.

More epochs should not be used as a substitute for a missing temporal mechanism.

23 Two-Month Research Roadmap

23.1 Weeks 1–2

- complete the corrected NVDS-lite baseline;
- implement aligned-local-only;
- implement forward-only causal BiDA propagation;
- validate flow, masks, state, causality, and gradients;
- run the one-seed baseline ladder.

23.2 Weeks 3–4

- promote decisive models to three seeds;
- evaluate SCARED geometry and temporal behaviour;
- benchmark runtime and memory;
- evaluate state reset and shuffled history;
- select one candidate architecture.

23.3 Weeks 5–6

- evaluate D4D by specimen, session, clip, and seed;
- evaluate SERV-CT geometry and safety;
- optionally pretrain using Hamlyn or EndoSLAM;
- add confidence or specularly masks if justified by ablation.

23.4 Weeks 7–8

- final ablations;
- scanline-over-time visualisations;
- qualitative surgical videos;
- bootstrap analysis;
- paper writing;
- supplementary and reproducibility audit.

24 Final Positioning

The strongest positioning is not:

We add temporal attention and a TGM loss to stereo disparity.

The stronger and more defensible positioning is:

Recommended Paper Positioning

BiDAStabilizer demonstrates that aligned local disparity evidence and bidirectional propagated state can stabilise framewise stereo predictions.

However, its use of future frames and unconstrained residual correction makes it unsuitable for online surgical robotics.

ARGOS studies whether this principle can be transformed into a causal, forward-only, plug-and-play, metric-safe temporal stereo refiner for deformable surgical scenes, and evaluates whether the resulting temporal gains survive out-of-distribution transfer without harming geometry.

The intended contribution is therefore the synthesis of:

plug-and-play stereo refinement+target-to-source alignment+causal state propagation+bounded metric correction

25 Verification Notes

The main BiDAStabilizer technical details used in this document were verified against:

- arXiv:2409.20283v2;
- the arXiv HTML full-text rendering;
- the IEEE TPAMI publication record;
- the BiDAVideo project page;
- the ECCV conference version of BiDAStereo.

Verified properties include:

- approximately 0.7 M trainable parameters;
- frozen image-based stereo backbone;

- RAFT optical flow;
- local aligned disparity feature extraction;
- forward and backward hidden-state propagation;
- convolutional Propagation Encoder;
- residual output;
- sequence length $T = 8$;
- 8 K training iterations;
- AdamW with learning rate 4×10^{-4} ;
- temporal consistency weight $\lambda = 0.2$;
- bidirectional offline inference;
- component and propagation ablations.

The exact hidden-state channel count and the BiDAStabilizer-specific training batch size were not explicitly identified in the reviewed primary source and should not be invented.

References

- [1] Y. Wang, M. Shi, J. Li, Z. Huang, Z. Cao, J. Zhang, K. Xian, and G. Lin. Neural Video Depth Stabilizer. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9466–9476, 2023.
- [2] Y. Wang et al. NVDS+: Towards Efficient and Versatile Neural Stabilizer for Video Depth Estimation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 47(2):1157–1173, 2024.
- [3] Z. Chen et al. Video Depth Anything: Consistent Depth Estimation for Super-Long Videos. In *CVPR*, 2025.
- [4] G. Chou et al. FlashDepth: Real-Time Streaming Video Depth Estimation at 2K Resolution. In *ICCV*, 2025.
- [5] N. Karaev, I. Rocco, B. Graham, N. Neverova, A. Vedaldi, and C. Rupprecht. DynamicStereo: Consistent Dynamic Depth from Stereo Videos. In *CVPR*, 2023.
- [6] Y. Zhang et al. TemporalStereo: Efficient Spatial-Temporal Stereo Matching Network. In *IROS*, 2023.
- [7] J. Zeng et al. Temporally Consistent Stereo Matching. In *ECCV*, 2024.
- [8] J. Jing, Y. Mao, A. Qiu, and K. Mikolajczyk. Match-Stereo-Videos: Bidirectional Alignment for Consistent Dynamic Stereo Matching. In *ECCV*, 2024.
- [9] J. Jing, Y. Mao, A. Qiu, and K. Mikolajczyk. Match Stereo Videos via Bidirectional Alignment. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 48(12):9337–9353, 2026. DOI: <https://doi.org/10.1109/TPAMI.2026.3679033>.
- [10] J. Jing et al. Stereo Any Video: Temporally Consistent Stereo Matching. In *ICCV*, 2025.
- [11] C. Wang et al. PPMStereo: Pick-and-Play Memory Construction for Consistent Dynamic Stereo Matching. In *NeurIPS*, 2025.

- [12] H. Xu, C. Xu, and S. Giannarou. StereoDiffusion: Temporally Consistent Stereo Depth Estimation with Diffusion Models. In *MICCAI*, 2024.
- [13] B. Cui et al. Surgical-DINO: Adapter Learning of Foundation Models for Depth Estimation in Endoscopic Surgery. In *IPCAI / IJCARS*, 2024.
- [14] Z. Zhou, C. Yang, P. Yang, X. Yang, and W. Shen. EndoDAV: Depth Any Video in Endoscopy with Spatiotemporal Accuracy. In *MICCAI*, 2025.
- [15] H. Li et al. EndoStreamDepth: Temporally Consistent Monocular Depth Estimation for Endoscopic Video Streams. In *MIDL*, PMLR volume 315, pages 1697–1721, 2026.
- [16] C. Budd and T. Vercauteren. Transferring Relative Monocular Depth to Surgical Vision with Temporal Consistency. In *MICCAI*, 2024.
- [17] H. Wang et al. StereoMamba: Real-Time and Robust Intraoperative Stereo Disparity Estimation. *arXiv:2504.17401*, 2025.
- [18] M. Allan et al. Stereo Correspondence and Reconstruction of Endoscopic Data Challenge. *arXiv:2101.01133*, 2021.
- [19] R. Docea et al. The Dresden Dataset for 4D Reconstruction of Non-Rigid Abdominal Surgical Scenes. *arXiv:2603.02985*, 2026.
- [20] P. Edwards et al. SERV-CT: A Disparity Dataset from CT for Validation of Endoscopic 3D Reconstruction. *Medical Image Analysis*, 77:102302, 2022.
- [21] K. Ozyoruk et al. EndoSLAM Dataset and an Unsupervised Monocular Visual Odometry and Depth Estimation Approach for Endoscopic Videos. *Medical Image Analysis*, 71:102058, 2021.
- [22] M. Hayoz et al. StereoMIS: Stereo Video Dataset for Robotic Surgery. Zenodo, 2023. <https://zenodo.org/records/8154924>.
- [23] Z. Teed and J. Deng. RAFT: Recurrent All-Pairs Field Transforms for Optical Flow. In *ECCV*, 2020.
- [24] G. Xu, X. Wang, X. Ding, and X. Yang. Iterative Geometry Encoding Volume for Stereo Matching. In *CVPR*, 2023.